Functional Worst-Risk Minimization in Non-Linear Models

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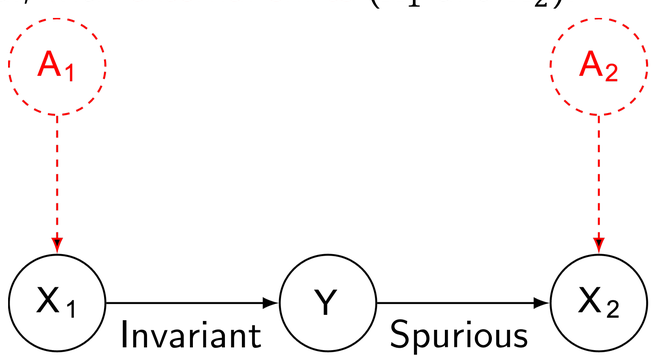
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Goal: Minimizing worst-case risk in non-linear functional systems by bounding errors under shifts

Theoretical Framework

1. Causality

In a **Functional System** example, Y is determined by the predictor variable X_1 and subsequently generates the variable X_2. This system is observed in 2 distinct environments, the **observational** and **interventional**, with external shifts (A_1 and A_2).



- In this setting, **causality** is defined by invariance of the risk.
- The true causal mechanism ($X_1 \to Y$) is the ONLY relationship that remains **invariant** across environments.

$$R_O(\beta_{causal}) = R_A(\beta_{causal})$$

- **Objective**: out-of-sample generalization, build a model that makes accurate predictions in unseen environments.

2. Past Work

Kennerberg and Wit (2025) introduced **functional worst-risk minimization** for linear relationships. This thesis extends their model to **non-linear functional relationships** by changing the operator definition from:

$(Y, X) = \mathcal{T}(Y, X) + \epsilon$ to $(Y, X) = \mathcal{T}_{\mathcal{NL}}(Y, X) + \epsilon$

Where the non-linear operator is defined as the **functional additive model** from **Fan et al. (2015)**

$$[\mathcal{T}_{\mathcal{NL}}(X)]_t = \sum_{j=1}^{p} g_j\left(\int_{T_1}^{T_2} \beta_j(t,\tau) X_\tau(j) d\tau\right)$$

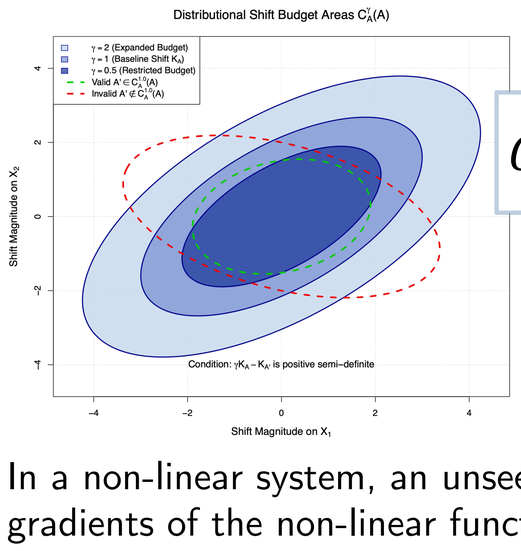
3. Main Result

Because of distributive failure, the exact **worst-risk decomposition** is no more possible: $g(\langle\beta, X + A\rangle) \neq g(\langle\beta, X\rangle) + g(\langle\beta, A\rangle)$

Instead of exact decomposition, we evaluate the worst-case scenario by taking the supremum of an unknown risk under unseen shifts.

$$\sup_{A' \in C_{\mathcal{A}}^{\gamma}(A)} R_{A'}(\mathcal{M}) \leq R_O(\mathcal{M}) + \gamma \cdot \Omega(K_A, \nabla\mathcal{M}) + \mathcal{O}\left(L_{lip}^2\right)$$

where $R_0(\mathcal{M})$is the baseline observational risk, Ω represents the specific sensitivity penalty of the predictor evaluated against the observed shift covariance K_A, and $L_{lip}$is the Lipschitz constant of the non-linear link function g.



$$C_{\mathcal{A}}^{\gamma}(A) = \left\{A' \in \mathcal{A} : \begin{matrix}\gamma K_A - K_{A'} \\ \text{is p.s.d.}\end{matrix}\right\}$$

This formula sets the boundary for allowable future shifts (A').

In a non-linear system, an unseen shift can interact with the steep gradients of the non-linear function, sending prediction error toward infinity. Our **main result** proves that this error can be **strictly contained** by establishing a tight **upper bound** for the **worst-case risk**, formulated as a **computable function of the two observed environments**.

4. Identifiability

Whilst on the population level the constraint was a DAG, for the estimation we additionally introduce:

$$\left|\left|\beta\right|\right|_2 = 1 \quad \text{and} \quad \mathbb{E}\left[g_j(u)\right] = 0$$

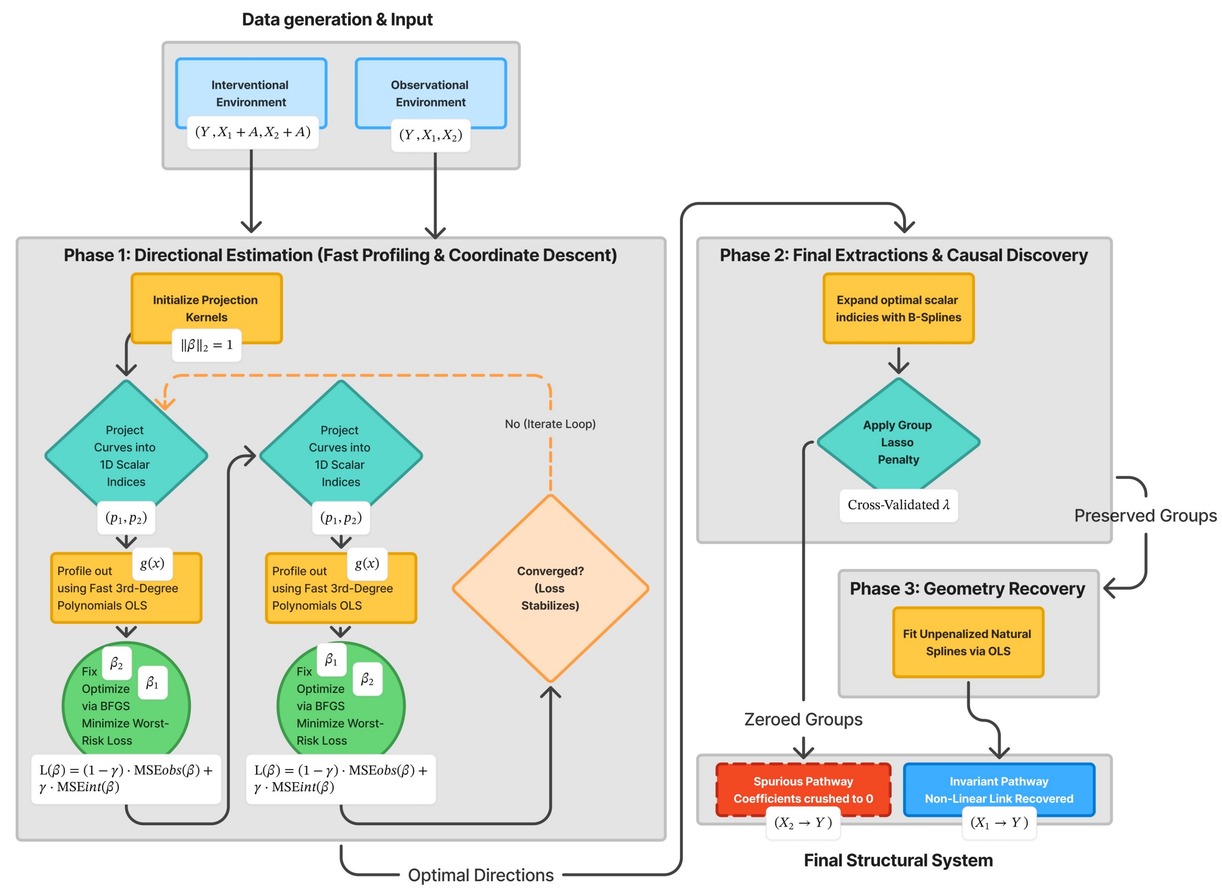
Empirical Part

5. Estimation

To estimate causal links across 2 environments without direct risk decomposition, we utilize an **empirical worst-risk formulation**:

$$\mathcal{L}(\beta, g) = (1 - \gamma) \cdot MSE_{Obs}(\beta, g) + \gamma \cdot MSE_{Int}(\beta, g)$$

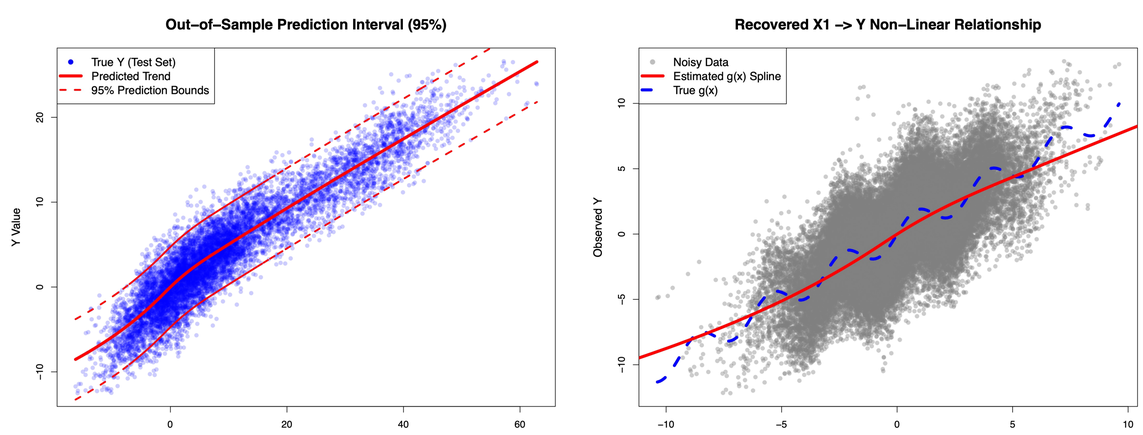
Using the **hyperparameter** $\gamma \in [0,1]$ to manage the **trade-off** between in-sample risk and out-of-sample risk.



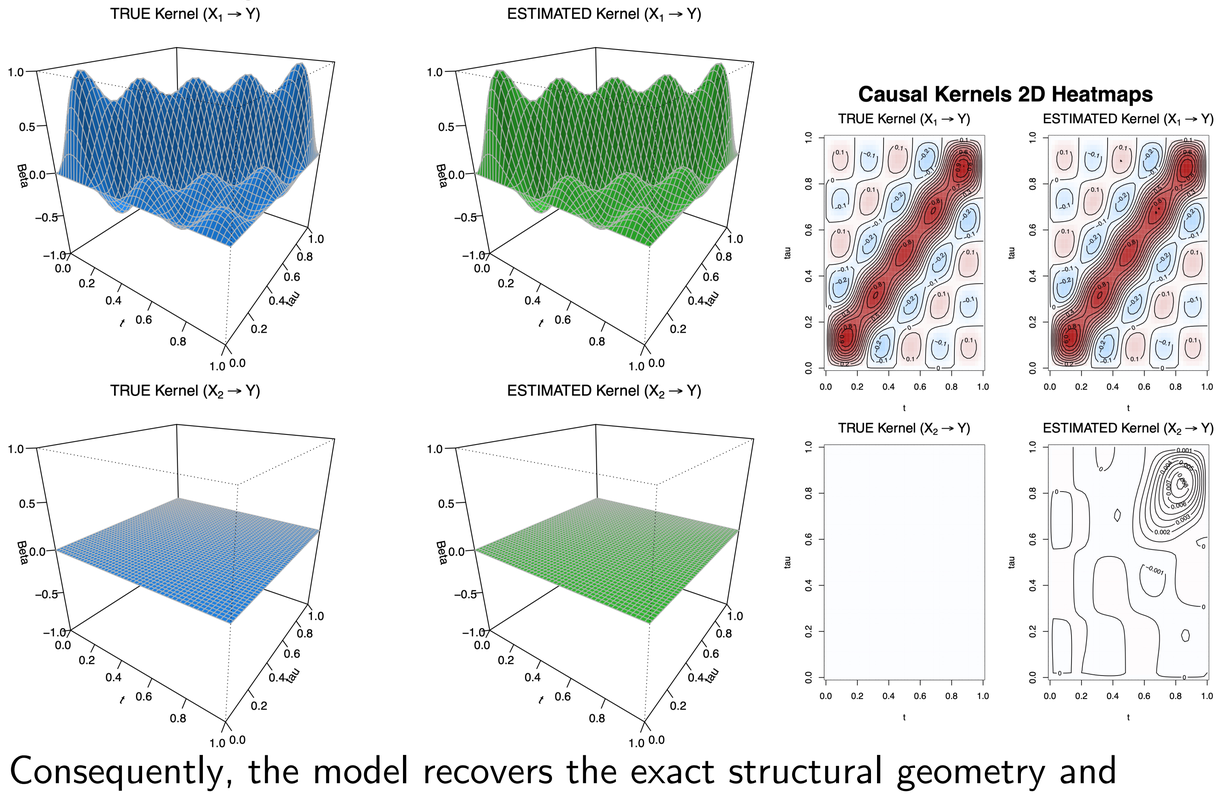
- Phase 1: **Alternating block coordinate descent** and fast **polynomial profiling** (lock in the projection kernels)
- Phase 2: Cross-validated **Group Lasso** to drop false variables
- Phase 3: Fitting an unpenalized **natural spline** that reveals the true non-linear effect

6. Empirical Results

We can observe how the splines were able to fit the underlying non-linear link function:



The underlying β kernel landscape is also closely estimated despite the task having a non-convex local optimization issue:



Consequently, the model recovers the exact structural geometry and maintains robust **predictive accuracy** even under severe **interventional shifts**.